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MOTHER-INFANT SEPARATION IN MONKEYS*

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THE problem of mother-child separation and the mechanisms underlying the disturbance it produces in the child have been discussed by Bowlby (1960, 1961), who has reviewed the literature and presented a theoretical position concerning separation anxiety, grief, and mourning in children and infants. In the later paper he outlined six theories of separation and concluded that separation anxiety results from activation of the component instinctual response systems, which are the base of the infant's attachments to the mother, in a situation in which the familiar object is not available. The effects on the child occur in three phases: protest, despair, and detachment, the first stemming from anxiety, the second from grief and mourning, and the third from defensive reactions.

If the mechanisms producing Bowlby's separation syndrome are as biologically basic as the activation of "basic component instinctual responses", a similar syndrome should be produced in infant monkeys following maternal separation. There is great similarity in the variables which Bowlby (1958) has described in his Component Instinctual Response theory and the variables producing the infant monkey's tie to its mother. Although Harlow would attach more importance to clinging and somewhat less importance to sucking than Bowlby does, there is little fundamental difference (Harlow, 1960; Harlow and Zimmermann, 1959). It is clear that clinging, which results in contact comfort, is a stronger and more persisting variable than nursing in the monkey, and any such differences in the importance of the components for monkey and man may be a function of the macaque monkey's maturational acceleration. The variable, following (visual and auditory responsiveness to the mother), has also been identified in studies concerning infant attachments of both species to their mothers (Bowlby, 1958; Harlow, in press), the responses appearing earlier in monkey than in man and being far stronger in the monkey's early infancy than in man's, probably because of the monkey's maturational acceleration.

The number of quantitative investigations on maternal separation in non-human primates is limited. Spence (1937), in a study involving permanent separation of infant chimpanzees from their mothers at various ages, observed the mothers' emotional responses to the sight of their infants. Five of seven mothers showed continuing own-infant specific emotional responses. Violent attempts to regain contact with the infants dropped out very quickly, but differential recognition of their own

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as opposed to other infants was still evident after separations of 6 months or more.

Short-term effects of baby exchanges of less than an hour's duration in two pig-tailed macaque mother-infant pairs have been reported recently by Jensen and Tolman (1962). The mothers in this study exhibited infant specificity after separation, but the two infants, which were 5 and 7 months old, initially showed positive responses to the strange mother. However, these responses dropped out very quickly in the face of rejection or disregard. In the process of being separated, both infants and mothers showed emotional responses; the babies screamed almost continuously, and the mothers attacked their cages and attempted to escape. The mothers were ferocious toward attendants, and the babies resisted attendants by clinging to the mother or to any available object. When the infants were returned to their mothers, there were striking initial increases in infant-mother and mother-infant responsiveness.

Over 200 infant rhesus monkeys have been separated from their mothers in the first 12 hours of life at the University of Wisconsin Primate Laboratory. In virtually all cases a team of men is required to restrain the mother while removing the infant, and the activity is not without some risk. The emotional reactions of these mothers are brief, and no increased animosity toward laboratory personnel has been observed consequent to separation.

During the last two years, a number of informal separation studies of an exploratory nature have been conducted at the Wisconsin Primate Laboratory on 30- to 90-day-old rhesus infants raised with their own mothers. The effects of separations ranging from a few hours to a few days in duration indicated acute disturbance of both mother and infant during separation, and a marked increase in maternal protectiveness after return of the infant. In one case the mother, after a 3-day separation, kept her infant literally within reach for a period of more than a month. Following these preliminary observations a more formal study of mother-infant separation was initiated.

METHOD

Subjects

Two pairs of infant rhesus monkeys and their four mothers served as subjects. Pair 1, Male A58 and Female A59, were separated at 170 and 169 days of age, respectively, and Pair 2, Female A63 and Male A64, were separated at 207 and 206 days of age. All animals had previously been removed from their mothers within 12 hr of birth for weighing and identification tattooing, and then returned. Infant A63 had also been removed from its mother at 36 days of age for 3 days' treatment of an infected hand.

Apparatus

The apparatus, shown in Fig. 1, included two woven-wire living-cages 36 in. by 36 in. by 36 in. adjoining two woven-wire playpen units 30 in. by 30 in. by 30 in., separated by a removable wire-mesh panel. The infants had access to the playpen units through 3·5- by 5·0-in. openings between home cages and playpen units.

Each pair of infants and each pair of mothers lived in such a test apparatus throughout the entire experiment, except for one hour each week when the apparatus

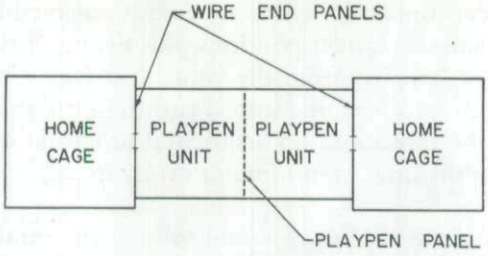


FIG. 1. Deprivation apparatus. See text for dimensional details.

was being cleaned. Pair 1 had lived in the apparatus since birth. Pair 2 had previously been housed in a slightly modified situation, and was placed in the standard separation apparatus three days prior to the first session of the pre-separation period.

Procedure

The experiment proper was divided into three periods of three weeks each, as follows: (1) Pre-separation: the infants continued to live with their mothers and essential base-line data were taken; (2) separation: the infants were separated from their mothers and lived together in the playpen; and (3) post-separation: the infants were allowed to return to their mothers and to live with them throughout the remainder of the experiment.

A day before the first separation test session the wire end-panels were replaced by clear Plexiglas end-panels of the same size. To permit the infants access to the playpen units, the plastic panels were fixed in a position 5 in. above the floor until the moment of experimental separation. When each infant entered a playpen unit in the first separation session, the appropriate Plexiglas end-panel was lowered, thus depriving the infant of all but auditory and visual contact with its mother. Just prior to the first post-separation test session the Plexiglas end-panels were replaced by the original wire end-panels. During pre-separation and post-separation the removable panel between playpens was in place except for the daily 30-min test sessions. During separation, however, this panel remained out at all times, allowing the infants free access to each other 24 hr a day.

Detailed observations were made of the behaviour of the animals 30 minutes a day, five days a week, using a symbol system and behaviour inventory developed in this laboratory. In some cases two or more individual categories of behaviour were combined for this presentation to provide a more general behaviour index. The list that follows includes only those items which occurred with reasonable frequency:

1. *Mother-Infant Responses*
   a. Cradle: support of the infant with one or both arms or with the legs while the infant is in contact with the mother.
   b. Visual contact: oriented "looking at" or visual exploration of the infant.
2. *Infant-Mother Responses*
   a. Clinging: clasping the mother's fur with hands or feet and so effecting some degree of body-weight support.
   b. Ventral contact: gross body contact with the ventral surface of the mother.

c. Support contact: contact in which the infant received body-weight support, including the ventral contact positions plus sitting or riding on the mother's back or head, or hanging from the ventral surface of the mother.
d. Non-specific contact: bumping into, brushing or leaning against, or otherwise touching the mother in an accidental or casual manner.
e. Nipple contact: sucking, mouthing, or orally manipulating the breast of the mother.
f. Visual contact: as in 1b, but oriented toward the mother.

3. *Infant-Infant Responses*
a. Non-contact play: activity characterized by running toward or away from the partner, involving either two distinct directional changes or a direct bee-line mock attack; this category also includes integrated play involving nipping and clasping and pulling, as well as non-contact play elements.
b. Threats: facial expressions involving ears back, mouth open, brow wrinkled, or some combination of these response elements oriented toward the other infant.
c. Approach: oriented movement of at least one body-length toward another monkey.
d. Withdrawal: oriented movement of at least one body-length away from another monkey.
e. Non-specific contact: as in 2d, but involving another infant.
f. Facilitation: imitative exploration or manipulation of objects immediately following similar manipulation by the other infant.

4. *Infant Non-social Responses*
a. Crying: soft, high-pitched vocalization (ooh, ooh).
b. Self-mouth: sucking or mouthing the self, usually involving a digit but scorable if directed to other parts of the body, e.g., the forearm.

The daily 30-min test session was divided into 120 15-sec periods. Three experienced observers participated, R.D. observing in each session and B.S. and E.H. alternating in a balanced order. Each observer recorded the behaviours of one infant-mother combination according to a predetermined order, so that subjects and testers were counterbalanced as far as possible. Behaviours were scored sequentially, but the maximum score for any one behaviour in any 15-sec period was 1, thus preventing overweighting of repetitive behaviours. Reliability sessions were conducted in a slightly different situation between E.H. and R.D. about 3 months prior to this study. All inter-tester product-moment reliability coefficients were above 0·90 for individual behaviour categories except for infant-mother non-specific contact ($r=0{\cdot}83$), infant-mother visual contact ($r=0{\cdot}64$), and the mouth-open component of infant-infant threats ($r=0{\cdot}78$). B.S. reached a training criterion of $r \geq 0{\cdot}85$ with E.H. and R.D. on all categories prior to testing.

RESULTS

The initial reactions of all mothers and all infants after separation indicated a high degree of emotional disturbance. The infants' behaviour generally included disoriented scampering, high-pitched screeching and crying, attempts to pass through the Plexiglas barriers, and huddling up against the barrier in close proximity to the

mother. The mothers showed an increase in barking vocalization and in threats directed toward the experimenter, but the mothers' emotional response appeared to be both less intense and of shorter duration than that of the infants.

Over-all effects before, during, and following separation were tested by a repeated-measures analysis of variance on each behaviour category listed except support contact. Results of the F and Duncan Multiple-Range tests (1955) show that two indices of infant behaviour, infant-mother visual contact (Fig. 2) and crying

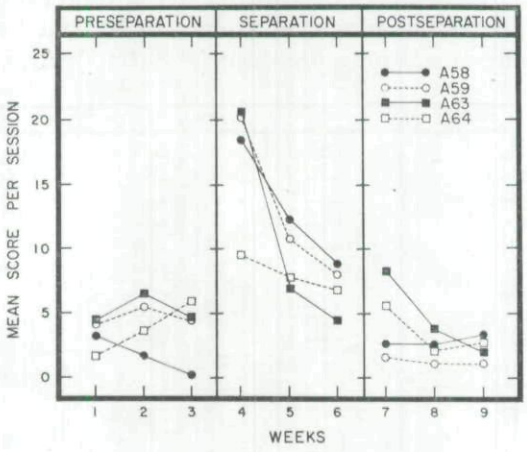


FIG. 2. Infant-mother visual contact as a function of weeks.

TABLE 1. INFANT CRYING VOCALIZATION: BEFORE, DURING, AND AFTER SEPARATION

Source	*d.f.*	MSS	*F*	*P*
Subjects	3	190·50		
Conditions	2	1335·62	8·23	<0·025
Ss × C	6	162·17		
Weeks/C	6	56·22	1·39	—
Ss × W/C	18	40·56		
INFANT-MOTHER VISUAL CONTACT: BEFORE, DURING, AND AFTER SEPARATION				
Subjects	3	3·99		
Conditions	2	241·97	17·79	< 0·05
Ss × C	6	13·60		
Weeks/C	6	39·42	7·56	< 0·01
Ss × W/C	18	5·21		

vocalization were significantly higher ($P < 0{\cdot}05$) during separation (see Table 1). Crying is consistently associated with disturbance in the infant monkey. Both responses showed a decrease as a function of weeks of separation, indicating either adaptation to the total situation or extinction of the specific responses. The mothers' data yielded no such clear-cut indication of disturbance common to the mothers as a group, and informal observations indicated that the mothers adapted earlier and

became more acceptant of the separation than the infants. Although we were reasonably certain that the mothers were disturbed for several days, no behaviour category indicating maternal disturbance changed significantly.

During the separation period, most complex infant-infant social behaviours exhibited a drastic decrease in frequency of occurrence. Non-contact play, illustrated in Fig. 3, showed a significant decrease (see Table 2); indeed, it was for all practical purposes obliterated. Threats, approaches, and withdrawals also decreased, but not significantly ($P > 0{\cdot}05$). As shown in Table 3, infant-infant imitative exploration of toys and apparatus increased significantly ($P < 0{\cdot}05$), but the increase in infant-infant non-specific contact was not significant ($P > 0{\cdot}05$). An increase in infant self-

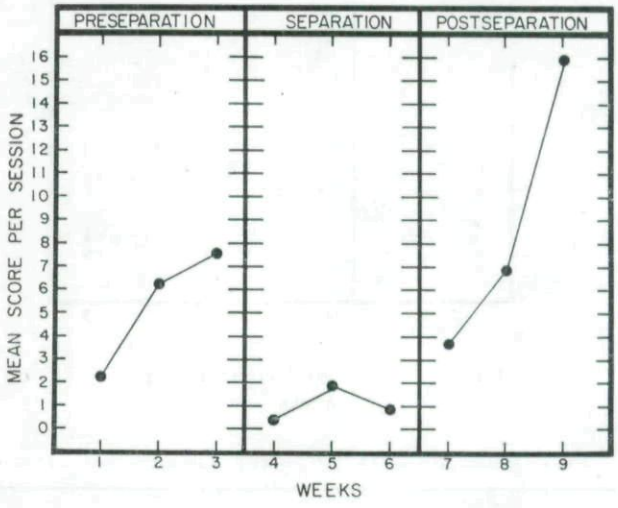


FIG. 3. Approach-withdrawal play as a function of weeks.

TABLE 2. INFANT-INFANT NON-CONTACT PLAY: BEFORE, DURING, AND AFTER SEPARATION

Source	*d.f.*	MSS	*F*	*P*
Subjects	3	135·48		
Conditions	2	193·02	8·99	<0·025
Ss × C	6	21·47		
Weeks/C	6	63·67	2·01	—
Ss × W/C	18	31·59		

TABLE 3. INFANT-INFANT IMITATE: BEFORE, DURING, AND AFTER SEPARATION

Source	*d.f.*	MSS	*F*	*P*
Subjects	3	0·41		
Conditions	2	5·03	11·03	<0·01
Ss × C	6	0·46		
Weeks/C	6	0·07	0·74	—
Ss × W/C	18	0·10		

mouthing and sucking during separation had been expected, but, instead, three of the four animals showed a decrease, A59 being the exception. A decrease in self-sucking with extreme emotional disturbance has been found previously by Bernstein and Mason (1962) and confirmed by Benjamin (1961). Intermediate levels of disturbance seem to bring about an increase in sucking. The experimenters had anticipated that separated infants might show the mutual ventral-ventral clinging seen in animals raised together from birth (Harlow, 1961), but no such increase was observed, despite the opportunities afforded by continuous association 24 hr a day.

Infant-directed behaviours by the mothers as a group were at a surprisingly low level throughout separation, but individual differences were large. The high variability is illustrated by the data on mother-infant visual contacts in Fig. 4. A comparison

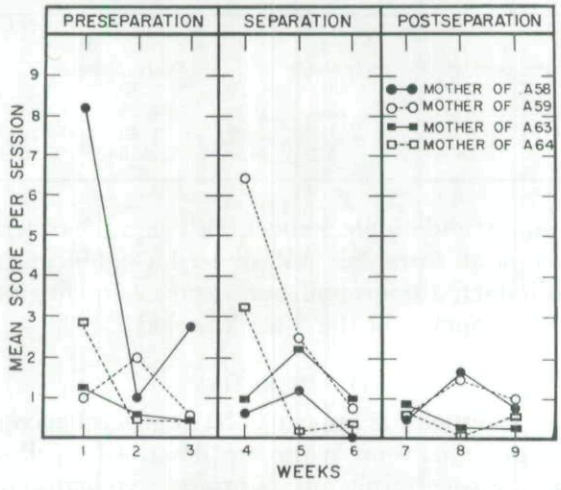


FIG. 4. Mother-infant visual contact as a function of weeks.

with Fig. 2, which shows infant-mother visual contact, further emphasizes the high variability of the mothers' behaviour compared with that of the infants.

An assessment of the effects of separation on initial post-separation behaviour for the group as a whole was made by comparing behaviour on the first day of post-separation with that on the last day of pre-separation. The results of *t* tests applied to these data are given in Table 4. Infant-mother clinging and mother-infant

TABLE 4. PRE- AND POST-SEPARATION INFANT-MOTHER AND MOTHER-INFANT BEHAVIOUR

Behaviour	Pre-separation mean (last day)	Post-separation mean (first day)	*P*
Mother-infant cradling	22·25	160·75	<0·05
Infant-mother clinging	2·75	89·00	<0·025
Infant-mother ventral contact	29·25	107·00	<0·10
Infant-mother nipple contact	24·75	16·25	<0·80
Infant-mother nonspecific contact	35·50	3·50	<0·01

cradling show a significant increase, and infant-mother non-specific contact shows a significant decrease. Ventral contact with the mother more than trebled, but the difference was not significant. Infant contact with the mothers' nipples showed a negligible average decrease.

Effects of separation throughout the three-week post-separation period were assessed in terms of individual animals. The results of simple between-within analyses of variance, comparing the frequencies of mother-directed behaviours of clinging, support contact, nipple contact and non-specific contact before and after separation for each infant, are presented in Table 5. Three infants showed a pattern of increased

TABLE 5. INDIVIDUAL INFANT-MOTHER RESPONSES: THREE-WEEK POST-SEPARATION *v.* THREE-WEEK PRE-SEPARATION PERIODS

	A58		A59		A63		A64	
Behaviour	% Increase	*P*	% Increase	*P*	% Increase	*P*	% Increase	P
Clinging	—30·2	<0·10	65·2	<0·20	58·8	>0·20	128·0	<0·05
Support contact	—57·1	<0·001	72·8	<0·05	29·5	>0·20	74·2	<0·05
Nipple contact	—58·1	<0·001	241·1	<0·001	45·7	>0·20	13·2	>0·20
Non-specific contact	28·8	<0·001	5·2	>0·20	—26·0	<0·001	25·2	>0·20

clinging, support contact, and nipple contact, although the differences did not reach statistical significance in all cases, but A58 showed a significant decrease in support contact and nipple contact, a non-significant decrease in clinging, and a significant increase in non-specific contact for the 3-week period.

DISCUSSION

The results of this investigation appear to be in general accord with expectations based upon the human separation syndrome described by Bowlby. The original response of the infants was certainly one of protest—violent and prolonged protest. This was followed by a period which can be appropriately described as despair. In the apparatus just prior to separation, as in our playpen studies generally, the infants invariably took every possible opportunity to engage in mutual play. The drastic depression of play following separation (see Fig. 3) is best explained by the trauma of separation. It might be noted that frequent daily informal observations through a one-way mirror in the wall separating the test room from the darkened hallway also revealed almost no play activity during the separation period. At least one monkey, No. A58, exhibited extreme disturbance during the first few separation days, showing definite signs of lack of sleep, and at times it seemed hardly able to stand or hold its eyes open. A58 also showed a decrease in thumb-sucking, as mentioned previously. Prior to separation, monkey A58 and its mother had shown significantly more contact scores than the other three pairs, suggesting a closer relationship for them than for the other pairs; A58's behaviour may therefore represent an extreme example of separation trauma.

We have not seen in our monkeys the full pattern constituting Bowlby's detachment phase, and except for monkey A58, we have not even observed a partial pattern. Three of the infants immediately attached to their mothers when separation was terminated, and showed some increase in all categories of mother-directed behaviours except non-specific contact throughout post-separation testing (Table 5). On the

other hand, A58 showed a decrease in most mother-directed behaviours except non-specific contact. Possibly this animal was showing some transient defence reactions against reattachment to the mother. We have not, after the reunion, observed signs of aggression against the mother such as those described by Robertson (1955) for children. Many factors could account for such differences: the period of maternal separation was relatively brief for the monkeys, the monkey mother had no part in the act of separation, strong aggressive patterns had not yet matured in the infant monkeys (these patterns appear in the second year), and the monkey mother is probably less acceptant of infant aggression than the human mother. However, it is more than possible that the two species differ in the degree to which resentment of the mother is formed and, in turn, expressed as aggression during separation and reunion. We believe that the human being, even the human child, is capable of more subtle and more profound behavioural disturbances than any non-human animal.

Bowlby (1961) has summarized various analytical theories concerning separation anxiety, and it is interesting to view them in terms of our separation data. Freud's signal theory cannot apply, since bodily needs of food, water, and even companionship were always available. Sexual capacity was certainly not threatened, since they were at an age when sexual responses to the mother had ended and shifted to playmates. Klein's depressive anxiety theory is not applicable, for even if infant monkeys were capable of believing that the mother disappeared because the infant ate her up, the mother monkey was continually present visually (Klein, 1935). Furthermore, we do not believe that our infants' distress can be explained in terms of persecutory anxiety, and seriously doubt that the rhesus infant possesses symbolic capability essential for such mechanisms. The fact that such theories seem untenable to explain separation anxiety in the monkey does not mean that they do not apply to the human being, but our data clearly show that separation anxiety can be produced in a primate infant by mechanisms as simple and direct as Bowlby subsumes in his concept of Primary Anxiety.

SUMMARY

The present study involved separation of four monkey mother-infant pairs for a 3-week period and measurement of the behaviour of the subjects before separation, during separation, and after reunion. All mothers and all infants showed emotional disturbance in response to separation, but the infants' disturbances were more intense and more enduring than those of the mothers. The findings were interpreted as being generally in accord with Bowlby's theory of primary separation anxiety as an explanatory principle for the basic primate separation mechanisms.

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